

**$\Delta$  vs frozen (good-oriented, value + 95% CI) · VANILLA-CONT-SSL block / SURGERY block · SCORE = mean per-metric-normalized (0-1) · BLUE BOX + big  $\Delta$  = best arm per metric**  
**WINNER row (champion duel — S=surgery V=vanilla-cont-SSL ==tie · tie = paired 95% CI overlaps 0): surgery 4 · vanilla-cont-SSL 1 · tie 4 → SURGERY**

|                                    | action_top1 ↑ better<br>classify motion class<br>from frozen features | motion_cos ↑ better<br>same-motion clips cluster<br>vs different motions | taxonomy_f1 ↑ better<br>read scene attributes<br>from features | future_mse ↓ better<br>predictor guesses<br>masked next-frame tokens | rollout ↓ better<br>error growth over<br>multi-step rollout | causal ↓ better<br>predict 2nd half<br>from 1st half | tdlst ↓ better<br>accuracy decay vs<br>prediction horizon | teacher_free ↓ better<br>free-run minus<br>teacher-forced gap | maskratio ↓ better<br>degradation under<br>heavier masking | SCORE<br>(norm 0-1) |
|------------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------------|----------------------------------------------------------------|----------------------------------------------------------------------|-------------------------------------------------------------|------------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------|------------------------------------------------------------|---------------------|
| ViT-G vanilla<br>continual-SSL     | +8.55<br>[+6.14,<br>+10.74]                                           | +0.14<br>[+0.14,<br>+0.15]                                               | -0.01<br>[-0.02,<br>+0.01]                                     | +0.02<br>[+0.02,<br>+0.03]                                           | 0<br>[0,<br>0]                                              | +0.02<br>[+0.02,<br>+0.03]                           | 0<br>[0,<br>0]                                            | +0.01<br>[+0.01,<br>+0.01]                                    | +0.01<br>[+0.01,<br>+0.01]                                 | 0.58                |
| ViT-G surgery<br>raw encoder       | +6.41<br>[+3.95,<br>+8.71]                                            | +0.12<br>[+0.12,<br>+0.13]                                               | -0.01<br>[-0.02,<br>+0.01]                                     | +0.06<br>[+0.06,<br>+0.06]                                           | 0<br>[0,<br>0]                                              | +0.06<br>[+0.05,<br>+0.06]                           | 0<br>[0,<br>0]                                            | 0<br>[0,<br>0]                                                | +0.02<br>[+0.02,<br>+0.02]                                 | 0.63                |
| ViT-G surgery<br>3stage_DI encoder | +4.88<br>[+2.41,<br>+7.34]                                            | +0.11<br>[+0.11,<br>+0.11]                                               | -0.01<br>[-0.02,<br>+0.01]                                     | +0.06<br>[+0.06,<br>+0.06]                                           | 0<br>[0,<br>0]                                              | +0.05<br>[+0.05,<br>+0.05]                           | 0<br>[0,<br>0]                                            | 0<br>[0,<br>0]                                                | +0.02<br>[+0.01,<br>+0.02]                                 | 0.58                |
| ViT-G surgery<br>3stage_DI head    | +5.70<br>[+3.51,<br>+7.84]                                            | +0.12<br>[+0.11,<br>+0.12]                                               | -0.01<br>[-0.03,<br>+0.01]                                     | +0.02<br>[+0.02,<br>+0.03]                                           | 0<br>[0,<br>0]                                              | +0.02<br>[+0.02,<br>+0.03]                           | 0<br>[0,<br>0]                                            | +0.01<br>[+0.01,<br>+0.01]                                    | +0.01<br>[+0.01,<br>+0.01]                                 | 0.42                |
| ViT-G surgical<br>autorgn encoder  |                                                                       | +0.18<br>[+0.17,<br>+0.18]                                               |                                                                | +0.03<br>[+0.03,<br>+0.03]                                           | 0<br>[0,<br>0]                                              | +0.03<br>[+0.03,<br>+0.03]                           | 0<br>[0,<br>0]                                            | 0<br>[0,<br>0]                                                | +0.01<br>[+0.01,<br>+0.01]                                 | 0.38                |
| ViT-G surgery<br>noDI encoder      | +6.30<br>[+4.00,<br>+8.60]                                            | +0.12<br>[+0.11,<br>+0.12]                                               | -0.01<br>[-0.03,<br>0]                                         | +0.06<br>[+0.06,<br>+0.06]                                           | 0<br>[0,<br>0]                                              | +0.05<br>[+0.05,<br>+0.05]                           | 0<br>[0,<br>0]                                            | -0.01<br>[-0.01,<br>-0.01]                                    | +0.02<br>[+0.01,<br>+0.02]                                 | 0.37                |
| ViT-G surgery<br>noDI head         | +5.37<br>[+3.01,<br>+7.78]                                            | +0.13<br>[+0.12,<br>+0.13]                                               | -0.01<br>[-0.03,<br>0]                                         | +0.02<br>[+0.02,<br>+0.03]                                           | 0<br>[0,<br>0]                                              | +0.02<br>[+0.02,<br>+0.03]                           | 0<br>[0,<br>0]                                            | +0.01<br>[+0.01,<br>+0.01]                                    | +0.01<br>[+0.01,<br>+0.01]                                 | 0.41                |
| WINNER                             | V                                                                     | S                                                                        | =                                                              | S                                                                    | =                                                           | S                                                    | =                                                         | =                                                             | S                                                          |                     |